

Challenge in Bayesian inference for climate reconstruction

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We consider a stationary heat conduction problem on the unit square $\Omega = [0, 1]^2$, governed by the Laplace equation

$$\Delta u(x, y) = 0 \in \Omega,$$

with homogeneous Dirichlet boundary conditions on three sides and an unknown, spatially varying Dirichlet boundary condition $u(0, y) = f(y)$ on the left boundary. The goal is to infer the unknown boundary function $f(y)$ from noisy temperature measurements taken at interior locations. Interior measurements are assumed to be related to the solution via

$$u_{\text{meas}} = \Phi(\alpha) + \eta,$$

where $\Phi(\alpha)$ denotes the forward model mapping boundary parameters to interior measurements and η the measurement noise.

The unknown boundary function is represented using a finite Radial Basis Function (RBF) expansion,

$$f(y) = \sum_{k=1}^K \alpha_k \exp\left(-\frac{(y - y_k)^2}{2\ell^2}\right),$$

where α_k are unknown coefficients, y_k uniformly distributed RBF centers along the boundary, and ℓ the RBF width. This representation enforces smoothness by construction. Given α , the forward model consists of solving the Laplace equation using a finite difference discretization with the prescribed boundary condition, and evaluating the solution at measurement locations using a Gaussian mollifier to model spatially averaged measurements.

Measurement noise is modeled as additive, independent Gaussian noise, leading to a Gaussian likelihood. This choice is motivated by the aggregation of multiple small and independent sources of experimental uncertainty, for which Gaussian noise is a natural approximation. Furthermore, the provided noise variance σ^2 supports this modeling assumption. A Gaussian prior is placed on the RBF coefficients, implying the belief that the boundary temperature is smooth and of moderate magnitude. The Gaussian prior acts as a regularization mechanism for the ill-posed inverse problem, penalizing unrealistically large boundary amplitudes while remaining weakly informative. The prior variance τ^2 is chosen on the scale of the observed data, allowing the measurements to dominate where informative. Finally, combining prior and likelihood yields the posterior distribution.

The posterior distribution is explored using a random-walk Metropolis–Hastings Markov Chain Monte Carlo (MCMC) sampler. This method is chosen for its simplicity, robustness, and suitability for non-Gaussian posteriors arising from nonlinear forward models. The sampler is initialized at $\alpha = 0$, corresponding to a homogeneous boundary condition, and the proposal step size is tuned to achieve a reasonable acceptance rate. Convergence is assessed using trace plots, running means, and autocorrelation analysis of representative coefficients. While the sampler exhibits autocorrelation due to correlated

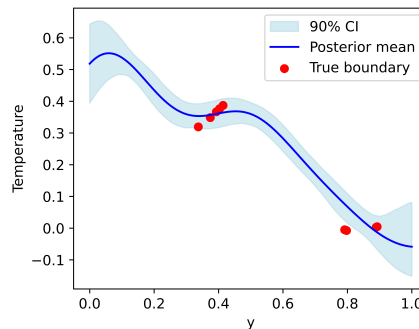


Figure 1: Bayesian reconstruction of the unknown boundary temperature. The solid line shows the posterior mean, the shaded region indicates the 90% credible interval, and red markers denote sparse samples of the true boundary provided for reference.

RBF coefficients and the smooth forward model, the running means after burn-in are stable and sufficient for uncertainty quantification.

Posterior samples are used to reconstruct the boundary function in a Bayesian sense. We report the posterior mean of $f(y)$ together with pointwise credible intervals, illustrating spatially varying uncertainty in the inferred boundary condition. Figure 1 shows the Bayesian reconstruction of the unknown boundary temperature $f(y)$.

The posterior mean captures the overall trend of the true boundary well, despite the limited and indirect information provided by interior measurements. Most reference samples lie within the credible interval, indicating that the inferred uncertainty is consistent with the observed discrepancy between model predictions and data. The width of the credible interval varies along the boundary. Regions with smaller uncertainty correspond to boundary segments that are more strongly informed by the interior measurements, while increased uncertainty near the endpoints reflects reduced identifiability due to weaker sensitivity of interior temperatures to those boundary regions.